Review 2 Commitment

1. Project Information

Project code	**FA26SE032**
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Review 2 date	29/10/2026
Reviewer 1	longt5
Reviewer 2	ThinhDP2
Version	08/11/2026
Date	08/11/2026

2. Final Registration Form

Final version	**Version 2.1 (Post-Review 2 Realigned Enterprise In-House Affiliate + Web & Mobile Scope)**
Date confirmed	08/11/2026
Record of Changes (BM03)	Form BM03 bundled in SEP490_FA26SE032_R2_Registration.pdf (6 realignments).

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	Multi-Role Auth, Registration/Reset Email OTP & Argon2id Security	Collaborator, Customer	Argon2id (>=64MB RAM) hashes passwords; 6-digit Email OTP verifies new registrations & password resets (>=5 wrong OTPs invalidate code); Login alerts via email; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	WF-01	Nguyễn Thành Thắng
F-02	Enterprise 5-Role RBAC & Privilege Escalation Guard	System	Strict RBAC enforces separation of duties across SYSTEM_ADMIN, SYSTEM_MANAGER, SHOP_MANAGER, COLLABORATOR, and CUSTOMER; public registration payload with role=SHOP_MANAGER is coerced to CUSTOMER.	WF-01	Nguyễn Thành Thắng
F-03	Internal Brand Staff & In-House Store Provisioning	System Admin	Only SYSTEM_ADMIN can create SHOP_MANAGER (bound to storeId & GHN satellite warehouse) and SYSTEM_MANAGER accounts; blocks public shop creation and logs immutable audit trail.	WF-01	Nguyễn Thành Thắng
F-04	Collaborator eKYC Dossier & Maker-Checker Verification [BR-01]	Collaborator, System Manager	Collaborator submits 12-digit CCCD (front/back photos), bank account & TikTok/FB/YouTube URL; System Manager independent audit verifies CCCD matches bank account name 100%; approval transitions KycStatus=VERIFIED, activates Bronze Tier (10% base rate), and assigns collaborator to in-house Store.	WF-01	Nguyễn Thành Thắng
F-05	2-Tier Product Variant Matrix Management & EAN-13 Barcodes	Shop Manager	Shop Manager creates 2-tier variants (Color/Size, Scent/Volume); validates unique EAN-13 barcodes across inventory; stock	WF-02	Phan Xuân Thịnh

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
			decrements atomically across warehouse reserves.		
F-06	MAP Floor-Price Guard & Product Price Moderation [BR-06]	Shop Manager, System Manager	System Manager configures SKU floorPrice; single/batch price promotions with proposedPrice < floorPrice return 400 Bad Request; discount activates only when status=ACTIVE within budget.	WF-02	Phan Xuân Thịnh
F-07	4-Axis AI Partner Matchmaking & Radar Scoring Engine	Shop Manager, Collaborator	AI engine scores verified Collaborators (0–100 across Category Relevance, Conversion Rate, Social Tier, Price Fit) in <=500ms; bilateral partnership invite transitions PENDING -> APPROVED.	WF-02	Phan Xuân Thịnh
F-08	Socket.IO Realtime Deal Chat & Media/Asset Sharing	Shop Manager, Collaborator	Real-time bi-directional chat allowing Shop Managers and Collaborators to negotiate terms and directly exchange product photos, review guidelines, and promotional assets with end-to-end integrity.	WF-02	Phan Xuân Thịnh
F-09	Autonomous Deal Negotiation (<=16%) & Admin Escalation (17–25%) [BR-08]	Shop Manager, System Admin	Counterparties exchange rich VIP Deal cards over Socket.IO chat; 10%–16% deals activate immediately on accept; accepted 17%–25% deals escalate to System Admin queue; proposals >25% fail with 400.	WF-02	Phan Xuân Thịnh
F-10	0 VND Sample Product Sponsorship & 14-Day Video Review SLA [BR-10]	Collaborator, Shop Manager	Verified Collaborators request 0 VND product samples (capped at maximum 3 active borrowed samples per store); Shop dispatches sample via GHN at 0 VND; receipt (RECEIVED) triggers 14-day SLA countdown.	WF-03	Nguyễn Đình Tuấn
F-11	Sample Video Review Verification & Overdue Borrowing Lock	Shop Manager, Collaborator	Collaborator submits review video URL within 14 days for Shop verification (COMPLETED); overdue samples past 14 days transition to OVERDUE and lock new sample requests.	WF-03	Nguyễn Đình Tuấn
F-12	Multi-Channel Attribution Asset Generation: Smart Links, QR & Coupons	Collaborator	Collaborator generates short referral link (/r/:shortCode), dynamic QR code, and exclusive deal goods coupon ([KOC][RATE], e.g., THAONHI16); public vouchers excluded.	WF-03	Nguyễn Đình Tuấn
F-13	Authenticated Checkout, Anti-Self-Referral Guard [BR-12] & PayOS VietQR	Customer, System	Checkout requires login while preserving 100% cart items; if Customer identity matches Collaborator by userId, phone, or bank account, commission is forced to 0%; encrypted PayOS VietQR webhook marks order PAID instantly.	WF-03	Nguyễn Đình Tuấn
F-14	Multi-Warehouse GHN Logistics Routing & Live Status Tracking [BR-13]	Shop Manager, GHN Courier	Shop Manager confirms order and dispatches GHN parcel from designated brand satellite warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED.	WF-04	Nguyễn Phú Quý
F-15	14-Day Return Claim Submission with Mandatory Unboxing Video [BR-14]	Customer	Customer initiates return request within 14 days of delivery; damaged goods claim requires photographic proof and mandatory unboxing video to isolate transit fault.	WF-04	Nguyễn Phú Quý
F-16	1-to-1 Damaged Item Exchange: 100% Commission & KPI Preserved [BR-14a]	Shop Manager, Customer	For transit damage (LOGISTICS_FAULT), Shop dispatches 1-to-1 replacement item at 0 VND; preserves 100% Collaborator commission and KPI progress; logs separate GHN compensation claim.	WF-04	Nguyễn Phú Quý
F-17	Automated Return Refund & Clawback Execution [BR-14b]	System, System Manager	Approved return refunds credit customer internal shopping wallet; simultaneously executes automated Clawback reversing 100% of order's held commissions to 0 VND and deducts refunded GMV from Deal KPI.	WF-04	Nguyễn Phú Quý
F-18	System Manager Platform Return Dispute Arbitration	System Manager	System Manager reviews escalated return disputes, inspects unboxing video and GHN tracking logs, and issues binding arbitration ruling (APPROVE_REFUND, APPROVE_EXCHANGE, UPHOLD_SHOP) protecting customers and collaborators.	WF-04	Nguyễn Phú Quý
F-19	Stage 1: Auto-Release Standard Remuneration	System	Automated reconciliation engine matures 10% base commission + Tier bonus (10%–13% total)	WF-05	Nguyễn Thành Thắng

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
	(10%+Tier) after 14D [BR-17]		from PENDING_14D_HOLD to availableBalance exactly 14 days after successful delivery without dispute.		
F-20	Stage 2: Exclusive Deal KPI Settlement (Escrow Release vs Forfeit) [BR-18]	System, Shop Manager	Holds VIP bonus in conditionalDealEscrowBalance until Deal hits 100% KPI (QUALIFIED) or expires without KPI (FORFEITED, keeping standard commission). Closed-month FR-09 milestone settles cleanly.	WF-05	Nguyễn Thành Thắng
F-21	Row-Locked Collaborator Withdrawal (SELECT FOR UPDATE) & 10% Flat PIT [BR-19,20]	Collaborator, System Manager	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200k VND) executes SELECT FOR UPDATE row-level lock on wallet; automatically withholds flat 10% personal income tax (FinancePosting).	WF-05	Nguyễn Thành Thắng
F-22	Disburse Napas 24/7 Sandbox & Executive Financial Ledger Reconciliation	System Admin, System	Disburses 90% net payout via Napas 24/7 Sandbox; automatically restores 100% funds on bank timeout (PAYOUT_REJECT_REFUND); updates real-time Executive Financial Ledger with zero reconciliation discrepancy.	WF-05	Nguyễn Thành Thắng
F-23	Immutable Security & Financial Audit Trail	System	100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete.	Platform / Audit	Nguyễn Thành Thắng
F-24	Category-Locked In-House Catalog & Vertical Isolation Guard	System	Backend service validates SKU creation against owning Store's locked primaryCategory (Cosmetics, Apparel, Nutrition, Smart Living); rejects cross-vertical listings.	Platform / Catalog	Phan Xuân Thịnh
F-25	Sample Sponsorship Quota Guard & Borrowing Limits	System	Background service enforces store-level sample inventory quotas and restricts Collaborators to a maximum of 3 concurrent active borrowed samples across in-house stores.	Platform / Samples	Nguyễn Đình Tuấn
F-26	30-Day Attribution Session & Fraud-Resilient Click Tracking [BR-11]	System	Referral link click sets 30-day AttributionSession cookie with Last-Click attribution precedence; Redis rate-limiting restricts spam clicks at 100 req/min/IP.	Platform / Attribution	Nguyễn Đình Tuấn
F-27	GHN Delivery Webhook Listener & Automated 14-Day Clock [BR-14]	System	Asynchronous webhook receiver processes GHN carrier status updates, transitions delivered shipments to DELIVERED atomically, and initializes the 14-day warranty return timer.	Platform / Shipping	Nguyễn Phú Quý
F-28	Monthly Milestone Bonus Settlement Engine (FR-09 Auxiliary)	System	Scheduled end-of-month cronjob aggregates clean GMV across 14-day matured orders, evaluates tier milestone rules, and credits milestone bonuses to eligible Collaborator wallets.	Platform / Commission	Nguyễn Thành Thắng
F-29	Automated KPI Forfeiture Resolution Engine	System	Automated scheduler evaluates expired Exclusive Deals, forfeits conditional deal escrow balances for unfulfilled targets, and returns escrow reservations to store budget.	Platform / Deals	Phan Xuân Thịnh
F-30	Executive Financial Ledger & Multi-Store Audit Reconciliation	System Admin, System	Enterprise financial balance sheet aggregating gross inflows, 14-day escrow liabilities, net payout disbursements, and 10% PIT tax obligations across all 4 in-house brand stores.	Platform / Finance	Nguyễn Thành Thắng

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-01	API & Redirect Latency	REST & Smart Link p95 Response Time	<= 200 ms for catalog queries and /r/:shortCode redirection	Autocannon benchmark (500 concurrent requests) & NestJS timing interceptor	Nguyễn Thành Thắng
NFR-02	Authentication & RBAC Security	Password Hash Cost & Role Isolation	Argon2id (>= 64 MB memory, 3 iterations); 100% cross-role calls blocked (403)	Jest security suites (auth-otp.spec.ts, rbac guards) across Web & Mobile endpoints	Nguyễn Thành Thắng

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-03	Web & Mobile Client Responsiveness	Web Initial Bundle / FCP & Mobile Bootstrap	Web gzipped entry <= 500 KB, FCP <= 1.5 s; Android /mobile/bootstrap <= 300 ms	Vite build bundle audit, Lighthouse CI, and Mobile contract/startup test suite	Phan Xuân Thịnh
NFR-04	Transactional & Inventory Integrity	Atomic Checkout & Return Rollback Rate	100% atomic rollback on stock exhaustion or invalid return state transition	Jest integration specs (orders.service.spec.ts, return-pickup.postgres.integration.spec.ts)	Nguyễn Phú Quý
NFR-05	Rate Limiting & Cache Resilience	Click Throttle & Redis Outage Failover	100 req/min/IP limit; 0 unhandled 500 errors during simulated Redis disconnect	CacheService LRU memory fallback unit tests & click-flood load script	Nguyễn Đình Tuấn
NFR-06	Financial Concurrency & Auditability	Double-Spend Prevention & Ledger Coverage	0 negative wallet balances under 20 parallel payouts; 100% mutations in AuditLog	Parallel SELECT ... FOR UPDATE payout concurrency test & FinancePosting assertions	Nguyễn Đình Tuấn

5. Out of Scope

Item	Reason
Open External Store Marketplace & Public Shop Registration	SCANMS governs external collaborators for in-house enterprise brand divisions; external external third parties cannot open public stores.
Generic Retail Voucher Stacking & Standalone Marketing Portals	Platform/freeship voucher stacking and standalone promotional portals are excluded to prevent margin cannibalization; only MAP-guarded ProductPricePromotion and KOC-bound attribution coupons are supported.
Factory Manufacturing, Supplier Procurement (ERP/AP) & Robotics WMS	Upstream OEM manufacturing, purchase orders, and warehouse hardware belong to corporate ERP; SCANMS manages sellable stock quotas and SKU floor prices.
Proprietary Courier Vehicles & Licensed Core Banking Rails	Physical parcel delivery and fiat banking rails are delegated to GHN Sandbox API, PayOS (order collection), and Napas 24/7 Sandbox API (collaborator payouts).
Standalone Live Video Streaming Platform & Core Media Production	Livestream is strictly an auxiliary creator promotion channel (+2% bonus capped at 25%); SCANMS does not provide an RTMP video encoding farm or open streaming platform. Livestream is excluded from the 5 core main workflows.

6. Workflow Commitments

6.1 Workflow WF-01

Workflow name	**WF-01: Onboard, Verify eKYC & Activate Sales Collaborators**
Owner (one member)	Nguyễn Thành Thắng
Supporting members	Phan Xuân Thịnh, Nguyễn Đình Tuấn
Actors	System Admin, Customer, System Manager, Collaborator
Trigger	Prospective collaborator initiates onboarding by verifying Email OTP and submitting an eKYC dossier (12-digit Citizen ID, beneficiary bank account, and social channel proofs); or System Admin provisions and assigns an in-house Shop Manager.
Business outcome	In-house Shop Manager bound to assigned brand store; verified Customer upgraded to Collaborator at Bronze (10%) tier with verified social channel and immutable AuditLog.
Main entity state path	User: isActive=true (SHOP_MANAGER/COLLABORATOR); KycStatus: UNVERIFIED -> VERIFIED / REJECTED; Tier: Bronze (10%)
Activity diagram (Report 4 section)	Report 4 Section 3.2.1 (Activity Diagram AD-WF01)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	Multi-Role Auth, Registration/Reset Email OTP & Argon2id Security	Argon2id (>=64MB RAM) hashes passwords; 6-digit Email OTP verifies new registrations & password resets (>=5 wrong OTPs invalidate code); Login alerts via email; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	TC_AUTH_01a, TC_AUTH_01b, TC_AUTH_01c
F-02	Enterprise 5-Role RBAC & Privilege Escalation Guard	Strict RBAC enforces separation of duties across SYSTEM_ADMIN, SYSTEM_MANAGER, SHOP_MANAGER, COLLABORATOR, and CUSTOMER; public registration payload with role=SHOP_MANAGER is coerced to CUSTOMER.	TC_AUTH_02a, TC_AUTH_02b, TC_AUTH_02c
F-03	Internal Brand Staff & In-House Store Provisioning	Only SYSTEM_ADMIN can create SHOP_MANAGER (bound to storeId & GHN satellite warehouse) and SYSTEM_MANAGER accounts; blocks public shop creation and logs immutable audit trail.	TC_AUTH_03a, TC_AUTH_03b, TC_AUTH_03c
F-04	Collaborator eKYC Dossier & Maker-Checker Verification [BR-01]	Collaborator submits 12-digit CCCD (front/back photos), bank account & TikTok/FB/YouTube URL; System Manager independent audit verifies CCCD matches bank account name 100%; approval transitions KycStatus=VERIFIED, activates Bronze Tier (10% base rate), and assigns collaborator to in-house Store.	TC_AUTH_04a, TC_AUTH_04b, TC_AUTH_04c

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	Main	Seeded System Admin, System Manager, and new Customer account; Sora Skincare store configured.	1. System Admin provisions in-house Shop Manager bound to Sora Skincare brand store. 2. Customer registers via Email OTP verification, then submits eKYC dossier (12-digit Citizen ID, bank account, and TikTok channel). 3. System Manager reviews and approves eKYC dossier with verified status.	Staff account active on Sora Skincare; Customer upgraded to COLLABORATOR with KycStatus=VERIFIED, initial tier Bronze (10%), and AuditLog recorded.	Nguyễn Thành Thắng

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01b	Exception	Unauthenticated client sends role=SHOP_MANAGER to /auth/register; KOC submits blurred CCCD or >5 wrong registration OTPs.	1. Attempt unauthorized registration requesting Shop Manager role. 2. Enter incorrect registration OTP 5 consecutive times. 3. System Manager reviews and rejects mismatched KYC citizen ID/bank dossier with mandatory reason.	Public signup coerced to CUSTOMER (privilege escalation blocked); registration OTP invalidated after 5 failures; KycStatus=REJECTED with feedback and resubmission enabled.	Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (>= 90%)	Nguyễn Thành Thắng
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Thành Thắng
State transition tests (>= 1 per transition)	2	Yes (100%)	Nguyễn Thành Thắng
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Thành Thắng
Total for this workflow	16	Yes (>= 90%)	Nguyễn Thành Thắng

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.2 Workflow WF-02

Workflow name	**WF-02: Moderate Product Variants, AI Match Collaborators & Negotiate Exclusive Deals**
Owner (one member)	Phan Xuân Thịnh
Supporting members	Nguyễn Thành Thắng, Nguyễn Phú Quý
Actors	Shop Manager, System Manager, Collaborator, System Admin
Trigger	Shop Manager manages 2-tier product variants with Floor Price guard (floorPrice), runs 4-Axis AI Partner Matchmaking, and initiates bi-directional Exclusive Deal negotiation via real-time Socket.IO chat.
Business outcome	Category-locked catalog approved, MAP floorPrice enforced, top KOC matched, and Exclusive Deal agreed (<=16% Shop Manager approval; >16% escalated to System Admin).
Main entity state path	ProductModerationStatus: DRAFT -> APPROVED / REJECTED; PricePromotionStatus: PENDING -> APPROVED -> ACTIVE; StoreCollaboratorStatus: PENDING -> APPROVED
Activity diagram (Report 4 section)	Report 4 Section 3.2.2 (Activity Diagram AD-WF02)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-05	2-Tier Product Variant Matrix Management & EAN-13 Barcodes	Shop Manager creates 2-tier variants (Color/Size, Scent/Volume); validates unique EAN-13 barcodes across inventory; stock decrements atomically across warehouse reserves.	TC_DEAL_01a, TC_DEAL_01b, TC_DEAL_01c
F-06	MAP Floor-Price Guard & Product Price Moderation [BR-06]	System Manager configures SKU floorPrice; single/batch price promotions with proposedPrice < floorPrice return 400 Bad Request; discount activates only when status=ACTIVE within budget.	TC_DEAL_02a, TC_DEAL_02b, TC_DEAL_02c
F-07	4-Axis AI Partner Matchmaking & Radar Scoring Engine	AI engine scores verified Collaborators (0–100 across Category Relevance, Conversion Rate, Social Tier, Price Fit) in <=500ms; bilateral partnership invite transitions PENDING -> APPROVED.	TC_DEAL_03a, TC_DEAL_03b, TC_DEAL_03c
F-08	Socket.IO Realtime Deal Chat & Media/Asset Sharing	Real-time bi-directional chat allowing Shop Managers and Collaborators to negotiate terms and directly exchange product photos, review guidelines, and promotional assets with end-to-end integrity.	TC_DEAL_04a, TC_DEAL_04b, TC_DEAL_04c
F-09	Autonomous Deal Negotiation (<=16%) & Admin Escalation (17–25%) [BR-08]	Counterparties exchange rich VIP Deal cards over Socket.IO chat; 10%–16% deals activate immediately on accept; accepted 17%–25% deals escalate to System Admin queue; proposals >25% fail with 400.	TC_DEAL_05a, TC_DEAL_05b, TC_DEAL_05c

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	Main	Sora Skincare (locked to Cosmetics), verified KOC profiles, and SKU floorPrice = 350,000 VND.	1. Shop Manager creates Vitamin C Serum (450k VND, EAN-13); System Manager approves. 2. Shop Manager proposes 390k VND promotion (>= floorPrice); System Manager approves. 3. Run 4-Axis AI Matchmaker, approve KOC collaboration, and counterparties exchange promotional guidelines and agree on exclusive campaign terms via real-time Chat.	Product APPROVED on storefront; promotion ACTIVE at 390k VND; AI returns ranked KOC radar scores; StoreCollaborator=APPROVED and agreement reached via Socket.IO chat.	Phan Xuân Thịnh
DS-02b	Exception	Shop Manager attempts cross-category SKU, duplicate EAN-13 barcode, or price drop below floorPrice (320k < 350k VND).	1. Submit Electronics category from Cosmetics store. 2. Submit duplicate EAN-13 barcode. 3. Propose promotion price 320,000 VND below 350,000 VND MAP floor.	All three requests rejected with 400 Bad Request; product and price remain unchanged, protecting brand vertical isolation and unit margin.	Phan Xuân Thịnh

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (>= 90%)	Phan Xuân Thịnh
End-to-end workflow test (>= 1)	1	Yes (100%)	Phan Xuân Thịnh
State transition tests (>= 1 per transition)	3	Yes (100%)	Phan Xuân Thịnh
NFR & security measurement tests (>= 1 per NFR)	2	Yes (100%)	Phan Xuân Thịnh
Total for this workflow	21	Yes (>= 90%)	Phan Xuân Thịnh

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.3 Workflow WF-03

Workflow name	**WF-03: Fulfill Experience Samples, Issue Marketing Tools & Capture Authenticated Attribution Orders**
Owner (one member)	Nguyễn Đình Tuấn
Supporting members	Phan Xuân Thịnh, Nguyễn Thành Thắng
Actors	Collaborator, Shop Manager, GHN Courier, Customer
Trigger	Collaborator requests 0 VND product sample (with 14-day review video SLA), generates Smart Link, Dynamic QR, and Exclusive KOL Coupon; Customer places authenticated zero-loss order with anti-self-referral verification.
Business outcome	Sample dispatched with 14-day SLA tracking; anti-self-referral guard blocks creator self-commission; multi-channel attribution captured with zero loss.
Main entity state path	SampleRequestStatus: PENDING -> APPROVED -> SHIPPED -> RECEIVED -> VIDEO_SUBMITTED -> COMPLETED / OVERDUE; AttributionSession: ACTIVE; OrderStatus: PENDING -> PAID
Activity diagram (Report 4 section)	Report 4 Section 3.2.3 (Activity Diagram AD-WF03)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-10	0 VND Sample Product Sponsorship & 14-Day Video Review SLA [BR-10]	Verified Collaborators request 0 VND product samples (capped at maximum 3 active borrowed samples per store); Shop dispatches sample via GHN at 0 VND; receipt (RECEIVED) triggers 14-day SLA countdown.	TC_SAM_01a, TC_SAM_01b, TC_SAM_01c
F-11	Sample Video Review Verification & Overdue Borrowing Lock	Collaborator submits review video URL within 14 days for Shop verification (COMPLETED); overdue samples past 14 days transition to OVERDUE and lock new sample requests.	TC_SAM_02a, TC_SAM_02b, TC_SAM_02c
F-12	Multi-Channel Attribution Asset Generation: Smart Links, QR & Coupons	Collaborator generates short referral link (/r/:shortCode), dynamic QR code, and exclusive deal goods coupon ([KOC][RATE], e.g., THAONHI16); public vouchers excluded.	TC_SAM_03a, TC_SAM_03b, TC_SAM_03c
F-13	Authenticated Checkout, Anti-Self-Referral Guard [BR-12] & PayOS VietQR	Checkout requires login while preserving 100% cart items; if Customer identity matches Collaborator by userId, phone, or bank account, commission is forced to 0%; encrypted PayOS VietQR webhook marks order PAID instantly.	TC_SAM_04a, TC_SAM_04b, TC_SAM_04c

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	Main	Approved Sora–KOC partnership on Web/Mobile; approved Vitamin C Serum with sample quota > 0.	1. KOC requests 0 VND sample for Vitamin C Serum -> Shop dispatches GHN at 0 VND. 2. KOC receives sample, submits video review URL within 14 days -> Shop approves COMPLETED. 3. KOC generates Smart Link (/r/:shortCode), Dynamic QR & Coupon (THAONHI16). 4. Customer signs in, preserves storefront cart, applies coupon, and pays via PayOS VietQR.	Sample reaches COMPLETED within 14-day SLA; order paid successfully via PayOS VietQR; 10% coupon discount applied and 100% attribution recorded to KOC.	Nguyễn Đình Tuấn
DS-03b	Exception	VIP Deal proposed at 28% (>25% cap); KOC misses 14-day sample review deadline; pinned SKU stock hits 0.	1. KOC misses 14-day review deadline and attempts new sample request. 2. Collaborator attempts self-referral purchase using own smart link with matching userId/phone/bank. 3. Client spams referral link exceeding 100 req/min.	Overdue sample transitions to OVERDUE and blocks new requests; Anti-Self-Referral Guard [BR-12] forces commission to 0%; Redis rate-limiter returns 429 Too Many Requests.	Nguyễn Đình Tuấn

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	9	Yes (>= 90%)	Nguyễn Đình Tuấn
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Đình Tuấn
State transition tests (>= 1 per transition)	4	Yes (100%)	Nguyễn Đình Tuấn
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Đình Tuấn
Total for this workflow	15	Yes (>= 90%)	Nguyễn Đình Tuấn

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.4 Workflow WF-04

Workflow name	**WF-04: Fulfill Multi-Warehouse Logistics, Resolve 14-Day Returns & Execute Clawback Reversal**
Owner (one member)	Nguyễn Phú Quý
Supporting members	Nguyễn Đình Tuấn, Phan Xuân Thịnh

Actors	Collaborator, Customer, Shop Manager, System Manager
Trigger	Collaborator shares Smart Link/QR and KOC Attribution Coupon; Customer checks out, receives GHN parcel, and files either a damaged-item Exchange or a Refund return.
Business outcome	Order fulfilled via dedicated brand GHN warehouse; delivery webhook triggers 14-day return window and hold; 1-for-1 EXCHANGE preserves 100% commission (+ carrier compensation claim) while REFUND claws back commission to 0 VND and deducts KPI GMV.
Main entity state path	OrderStatus: PENDING -> SHIPPING -> DELIVERED -> COMPLETED / RETURNED; ReturnRequestStatus: REQUESTED -> SHOP_APPROVED -> INSPECTING -> EXCHANGE_SHIPPED / REFUNDED -> COMPLETED; CommissionStatus: PENDING -> APPROVED / REVERSED
Activity diagram (Report 4 section)	Report 4 Section 3.2.4 (Activity Diagram AD-WF04)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-14	Multi-Warehouse GHN Logistics Routing & Live Status Tracking [BR-13]	Shop Manager confirms order and dispatches GHN parcel from designated brand satellite warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED.	TC_FUL_01a, TC_FUL_01b, TC_FUL_01c
F-15	14-Day Return Claim Submission with Mandatory Unboxing Video [BR-14]	Customer initiates return request within 14 days of delivery; damaged goods claim requires photographic proof and mandatory unboxing video to isolate transit fault.	TC_FUL_02a, TC_FUL_02b, TC_FUL_02c
F-16	1-to-1 Damaged Item Exchange: 100% Commission & KPI Preserved [BR-14a]	For transit damage (LOGISTICS_FAULT), Shop dispatches 1-to-1 replacement item at 0 VND; preserves 100% Collaborator commission and KPI progress; logs separate GHN compensation claim.	TC_FUL_03a, TC_FUL_03b, TC_FUL_03c
F-17	Automated Return Refund & Clawback Execution [BR-14b]	Approved return refunds credit customer internal shopping wallet; simultaneously executes automated Clawback reversing 100% of order's held commissions to 0 VND and deducts refunded GMV from Deal KPI.	TC_FUL_04a, TC_FUL_04b, TC_FUL_04c
F-18	System Manager Platform Return Dispute Arbitration	System Manager reviews escalated return disputes, inspects unboxing video and GHN tracking logs, and issues binding arbitration ruling (APPROVE_REFUND, APPROVE_EXCHANGE, UPHOLD_SHOP) protecting customers and collaborators.	TC_FUL_05a, TC_FUL_05b, TC_FUL_05c

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04a	Main	Verified KOC with active 16% VIP deal & KOC coupon (10% off); authenticated Customer; Sora GHN warehouse.	1. KOC generates smart affiliate link, dynamic QR & exclusive coupon. 2. Customer clicks link / applies KOC coupon, logs in preserving 100% cart, pays via PayOS/COD. 3. Shop creates GHN waybill -> DELIVERED. 4. On transit damage, Customer chooses EXCHANGE -> Shop ships 1-1 replacement & opens GHN claim.	Order attributed to KOC with pending commission recorded; replacement delivered (COMPLETED), 100% KOC commission preserved, and GHN compensation claim opened.	Nguyễn Phú Quý

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04b	Exception	KOC attempts self-purchase via own link/coupon; Customer requests full return & refund within 14-day window.	1. KOC checks out using own referral link/matching phone. 2. Separate Customer files REFUND return with unboxing video -> parcel returned & inspected -> refund approved.	Self-referral order sets commission rate = 0% and flags AiFraudService; refunded order credits Customer shoppingBalance and reverses KOC commission to REVERSED.	Nguyễn Phú Quý

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (>= 90%)	Nguyễn Phú Quý
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Phú Quý
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Phú Quý
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Phú Quý
Total for this workflow	17	Yes (>= 90%)	Nguyễn Phú Quý

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.5 Workflow WF-05

Workflow name	**WF-05: Settle Two-Stage Commission, Withhold 10% PIT & Disburse Payouts**
Owner (one member)	Nguyễn Thành Thắng
Supporting members	Nguyễn Đình Tuấn, Nguyễn Phú Quý
Actors	Shop Manager, Collaborator, System Manager, System Admin
Trigger	14-day return window elapses on delivered orders, closed-month milestone settlement evaluates, active Exclusive Deal achieves 100% KPI, and Collaborator requests bank withdrawal.
Business outcome	Stage 1 (10% + Tier%) unlocks after 14 days clean delivery; Stage 2 KPI differential unlocks upon 100% KPI achievement; 10% flat PIT withheld under DB row lock; net payout disbursed via Napas 24/7; audited in Executive Financial Ledger.
Main entity state path	CommissionStatus: PENDING -> APPROVED; DealBonusStatus: PENDING_KPI -> QUALIFIED / FORFEITED; PayoutStatus: PENDING -> PROCESSING -> APPROVED / REJECTED
Activity diagram (Report 4 section)	Report 4 Section 3.2.5 (Activity Diagram AD-WF05)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-19	Stage 1: Auto-Release Standard Remuneration (10%+Tier) after 14D [BR-17]	Automated reconciliation engine matures 10% base commission + Tier bonus (10%–13% total) from PENDING_14D_HOLD to availableBalance exactly 14 days after successful delivery without dispute.	TC_PAY_01a, TC_PAY_01b, TC_PAY_01c
F-20	Stage 2: Exclusive Deal KPI Settlement (Escrow Release vs Forfeit) [BR-18]	Holds VIP bonus in conditionalDealEscrowBalance until Deal hits 100% KPI (QUALIFIED) or expires without KPI (FORFEITED, keeping standard commission). Closed-month FR-09 milestone settles cleanly.	TC_PAY_02a, TC_PAY_02b, TC_PAY_02c
F-21	Row-Locked Collaborator Withdrawal (SELECT FOR UPDATE) & 10% Flat PIT [BR-19,20]	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200k VND) executes SELECT FOR UPDATE row-level lock on wallet; automatically withholds flat 10% personal income tax (FinancePosting).	TC_PAY_03a, TC_PAY_03b, TC_PAY_03c
F-22	Disburse Napas 24/7 Sandbox & Executive Financial Ledger Reconciliation	Disburses 90% net payout via Napas 24/7 Sandbox; automatically restores 100% funds on bank timeout (PAYOUT_REJECT_REFUND); updates real-time Executive Financial Ledger with zero reconciliation discrepancy.	TC_PAY_04a, TC_PAY_04b, TC_PAY_04c

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05a	Main	Delivered order past 14-day hold with no open return; closed-month clean GMV > 50M VND; verified KOC wallet.	1. Reconciliation job matures 14-day PENDING commission to availableBalance and qualifies KPI deal escrow. 2. Shop settles closed-month FR-09 milestone bonus & promotes KOC tier. 3. KOC requests 2,072,000 VND withdrawal -> System Manager approves via Napas 24/7 Sandbox.	Wallet debited once under SELECT ... FOR UPDATE; 207,200 VND (10% PIT) posted via FinancePosting; 1,864,800 VND net approved with SBX-... reference and AuditLog.	Nguyễn Thành Thắng
DS-05b	Exception	SHOP_MANAGER attempts withdrawal; KOC requests < 200,000 VND or fires parallel requests exceeding balance; Napas 24/7 Sandbox timeout.	1. Unauthorized role (Shop Manager) attempts balance withdrawal. 2. Submit 100,000 VND (<200k min) and 2 concurrent full-balance requests. 3. Simulate BIDV timeout/rejection.	SHOP_MANAGER blocked (403 Forbidden); <200k rejected (400); row lock allows at most 1 request (0 negative balance); rejected payout restores 100% funds via PAYOUT_REJECT_REFUND.	Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	12	Yes (>= 90%)	Nguyễn Thành Thắng
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Thành Thắng
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Thành Thắng
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Thành Thắng
Total for this workflow	17	Yes (>= 90%)	Nguyễn Thành Thắng

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

7. Individual Commitments

7.1 Member 1

Full name	**Nguyễn Thành Thắng**
Student ID	SE184251
Role in team	F-01..F-04, F-17..F-20

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-01: Onboard, Verify eKYC & Activate Sales Collaborators; WF-05: Settle Two-Stage Commission, Withhold Personal Income Tax & Disburse Payouts	Sole workflow owner
Mandatory features owned	WF-01: F-01 (Auth OTP), F-02 (5-Role RBAC), F-03 (Staff Provisioning), F-04 (eKYC Maker-Checker); WF-05: F-19 (10%+Tier 14D Release), F-20 (KPI Deal Escrow), F-21 (Row-Lock & 10% PIT), F-22 (Napas & Ledger); Platform: F-23 (AuditLog), F-28 (Milestone FR-09), F-30 (Executive Ledger) [11 features total]	Disjoint feature ownership
Modules / entities owned	AuthModule, UsersModule, StoresModule, KycModule, AuditModule, CommissionsModule, WalletsModule, PayoutsModule	Code & unit/E2E tests
Design decisions owned	D1: In-House Brand Provisioning & Shared Mobile/Web Bootstrap; D2: MAP Floor-Price Guard & KOC Attribution Coupon	BM01 Section 3
NFRs responsible	NFR-01 (API & Redirect Latency <= 200ms), NFR-02 (Argon2id & 5-Role RBAC Isolation)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	33 committed functional test cases	Defense 1.1
Of which automated in CI	33 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-01a, DS-01b, DS-05a, DS-05b	Defense 1.1
Required fixes owned	G1 (Sync SDD 69-model ERD & System Overview), G4 (Clean legacy deprecated stubs)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-01, WF-05	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified		

Member confirmation

Signature / date	**Confirmed by Nguyễn Thành Thắng (SE184251) · 08/11/2026**

7.2 Member 2

Full name	**Phan Xuân Thịnh**

Student ID	SE184527
Role in team	F-05..F-09

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-02: Moderate Products, Match AI Partners & Negotiate Exclusive Deals	Sole workflow owner
Mandatory features owned	WF-02: F-05 (Variant Matrix & EAN-13), F-06 (MAP Floor-Price Guard), F-07 (4-Axis AI Matchmaker), F-08 (Socket.IO Chat & Assets), F-09 (Autonomous Deal & Escalation Gate); Platform: F-24 (Category Lock Guard), F-29 (KPI Forfeiture Resolution Engine) [7 features total]	Disjoint feature ownership
Modules / entities owned	ProductsModule, PricingModule, AiModule, ChatModule	Code & unit/E2E tests
Design decisions owned	D2: MAP Floor-Price Guard & Category Lock; D3: Bi-Directional Exclusive Deal Negotiation via Socket.IO	BM01 Section 3
NFRs responsible	NFR-03 (Web Bundle <= 500KB / FCP <= 1.5s & Android Mobile Bootstrap <= 300ms)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	21 committed functional test cases	Defense 1.1
Of which automated in CI	21 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-02a, DS-02b	Defense 1.1
Required fixes owned	G2 (Mobile native checkout handoff & staging Android APK verification)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-02	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified		

Member confirmation

Signature / date	**Confirmed by Phan Xuân Thịnh (SE184527) · 08/11/2026**

7.3 Member 3

Full name	**Nguyễn Phú Quý**
Student ID	SE180104
Role in team	F-13..F-16

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-04: Execute Logistics, Resolve Dual-Path Returns & Reconcile Commission Clawback	Sole workflow owner

Type	Items (IDs / names)	Notes
Mandatory features owned	WF-04: F-14 (GHN Satellite Routing & Tracking), F-15 (14D Return Claim & Video), F-16 (1-to-1 Exchange 100% Fee Kept), F-17 (Refund & Clawback Reversal), F-18 (System Manager Dispute Arbitration); Platform: F-27 (GHN Delivery Webhook Listener & 14-Day Clock) [6 features total]	Disjoint feature ownership
Modules / entities owned	ShippingModule (GHN Express), ReturnsModule, OrdersModule	Code & unit/E2E tests
Design decisions owned	D4: 14-Day Return Window & Dual-Path Return Resolution (1-1 Exchange vs Refund Clawback)	BM01 Section 3
NFRs responsible	NFR-04 (100% Atomic Checkout & Return State Transition Rollback)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	17 committed functional test cases	Defense 1.1
Of which automated in CI	17 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-04a, DS-04b	Defense 1.1
Required fixes owned	0 open High defects (GHN compensation claim & dual-path return integration verified)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-04	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified		

Member confirmation

Signature / date	**Confirmed by Nguyễn Phú Quý (SE180104) · 08/11/2026**

7.4 Member 4

Full name	**Nguyễn Đình Tuấn**
Student ID	SE182540
Role in team	F-14..F-20 (Samples, Marketing Tools, Attribution & Secure Checkout)

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-03: Fulfill Experience Samples, Issue Marketing Tools & Capture Authenticated Attribution Orders	Sole workflow owner
Mandatory features owned	WF-03: F-10 (Sample 0 VND Sponsorship & 14D SLA), F-11 (Sample Video Verification & Overdue Lock), F-12 (Smart Link, Dynamic QR & Exclusive Coupon), F-13 (Authenticated Checkout & Anti-Self-Referral Guard); Platform: F-25 (Sample Quota Guard), F-26 (Attribution 30D Session & Redis Rate Limit) [6 features total]	Disjoint feature ownership
Modules / entities owned	SamplesModule, ReferralLinksModule, CouponsModule, CheckoutModule, OrdersModule, MobileModule & Expo Android App	Code & unit/E2E tests

Type	Items (IDs / names)	Notes
Design decisions owned	D1: Mobile Companion App & Smart Attribution Tools; D3: 14-Day Sample SLA Enforcement	BM01 Section 3
NFRs responsible	NFR-05 (100 req/min/IP Rate Limit & Redis Fallback), NFR-06 (Row-Lock Concurrency & 100% Ledger Auditability)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	15 committed functional test cases	Defense 1.1
Of which automated in CI	15 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-03a, DS-03b	Defense 1.1
Required fixes owned	G3 (Maintain Napas 24/7 Sandbox idempotency/inquiry adapter & FinancePosting ledger docs)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-03	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified		

Member confirmation

Signature / date	Confirmed by Nguyễn Đình Tuấn (SE182540) · 08/11/2026

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01: Staff Provisioning & eKYC	Nguyễn Thành Thắng	F-01..F-04 (4)	DS-01a / DS-01b	12	12
WF-02: Product Governance & Deals	Phan Xuân Thịnh	F-05..F-09 (5)	DS-02a / DS-02b	15	15
WF-03: Samples & Marketing Attribution	Nguyễn Đình Tuấn	F-10..F-13 (4)	DS-03a / DS-03b	12	12
WF-04: GHN Logistics & Dual-Path Returns	Nguyễn Phú Quý	F-14..F-18 (5)	DS-04a / DS-04b	15	15
WF-05: Two-Stage Payouts & 10% PIT	Nguyễn Thành Thắng	F-19..F-22 (4)	DS-05a / DS-05b	12	12
Platform & Background Services	System & Leads	F-23..F-30 (8)	Automated & Cronjobs	24	24
Total (deduplicated)	**All 4 members**	**30 features**	**10 main demo scenarios**	**114**	**114**

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Nguyễn Thành Thắng (Leader)	WF-01 & WF-05	11 features (8 WF + 3 Platform)	41	41	DS-01a, DS-01b, DS-05a, DS-05b	Oral presentation + Live demo
Phan Xuân Thịnh	WF-02	7 features (5 WF + 2 Platform)	27	27	DS-02a, DS-02b	Oral presentation + Live demo
Nguyễn Đình Tuấn	WF-03	6 features (4 WF + 2 Platform)	23	23	DS-03a, DS-03b	Oral presentation + Live demo
Nguyễn Phú Quý	WF-04	6 features (5 WF + 1 Platform)	23	23	DS-04a, DS-04b	Oral presentation + Live demo
Total (deduplicated)	**5 Main Workflows**	**30 features**	**114**	**114**	**10 scenarios**	**100% defense coverage**

8.3 Committed Numbers

Number of main workflows	5 (WF-01..WF-05; 1 sole owner per workflow)
Number of demo scenarios (main / exception)	30 (all with acceptance criteria, mapped to workflows)
Committed functional test cases	114 (minimum committed total across features, WFs, states, NFRs)
Percentage automated in CI	>= 90% (CI pipeline on review-2 tag)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	100% run end-to-end	100% (5/5 workflows run end-to-end on Cloud Staging & Mobile API)
Demo scenarios	100% main; >= 80% exception	100% main (5/5) and 100% exception (5/5) reproducible via seed script
Committed test cases pass rate	>= 90% at Defense 1.1	>= 90% of committed 86 test cases executed automatically in CI
Automated in CI	>= 50% of committed total	>= 90% of committed 81 test cases executed automatically in CI
Open Critical / High defects	0 open Critical or High defects	0 open Critical or High defects at Defense 1.1
SRS/SDD & Self-Review alignment	No unexplained High findings	All BM02 Planned items (SR-01..SR-04) closed with commit links
Individual ownership	Every member demos owned workflow	Each of the 4 members demos and defends their owned workflow(s) live

9. Required Fixes

ID	Item	Related workflow	Owner	Due
FX-01 (SR-01)	Regenerate Report 4 Logical ERD & Data Dictionary to match all 69 Prisma models and 5 UserRole values	WF-01, WF-02	Nguyễn Thành Thắng	20/11/2026
FX-02 (SR-02)	Update Report 4 System Overview Diagram & Tech Table (React/Vite, Expo Android, @nestjs/schedule, BIDV/PayOS/GHN)	WF-01	Nguyễn Thành Thắng	20/11/2026
FX-03 (SR-03)	Hide legacy Mobile menu stub and mark deprecated database table in SDD after scope cleanup	WF-02	Nguyễn Thành Thắng	20/11/2026
FX-04 (SR-04)	Verify Web-checkout handoff, staging Cloud deployment, and 5-role Android APK E2E test run	WF-03	Phan Xuân Thịnh	25/11/2026

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1	longt5		08/11/2026
Reviewer 2	ThinhDP2		08/11/2026
Supervisor	ThS. Tôn Thất Hoàng Minh		08/11/2026
Team leader	Nguyễn Thành Thắng		08/11/2026